

Data Hall Cooling Optimisation Report

Tampines Data Centre 1 · Data Centre · Singapore

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Prepared by	Data Centre Engineering

1. Summary

Tampines Data Centre 1 (TDC1) is an 8 MW colocation facility with three data halls and a current IT load of 5.1 MW. In 2024 its annualised power usage effectiveness (PUE) was 1.68, well above the portfolio target of 1.50, and cooling accounted for roughly 70% of the non-IT energy. A cooling optimisation programme in the data halls reduced PUE to 1.52 and cooling energy by 34%.

2. Baseline findings

- CRAH units supplied air at 18°C with fans fixed at 100% speed, far colder than the IT equipment required.
- A thermal survey and CFD study found 42% bypass airflow and hot-air recirculation over rack tops: rack inlet temperatures ranged from 19°C to 27°C within the same row.
- Blanking panels were missing in 30% of racks and cable cut-outs were unsealed.
- Chilled water was supplied at 7°C, the same as a typical office plant.

3. Measures implemented

- Hot-aisle containment in all three data halls, with blanking panels and brush grommets sealing every opening.
- CRAH supply air temperature raised in 1°C steps from 18°C to 24°C, keeping every rack inlet inside the ASHRAE TC9.9 recommended envelope of 18-27°C.
- Three temperature sensors per rack (top, middle and bottom) now drive CRAH fan speed through variable speed drives; fans typically run at 55-70%.
- Chilled-water supply temperature raised from 7°C to 12°C, which improves chiller efficiency and removes most unnecessary dehumidification.
- Customer SLAs were reviewed before each step, and every change had a documented rollback.

4. Results

KPI	2024	After programme
PUE (annualised)	1.68	1.52
Cooling energy (GWh/yr)	21.2	14.0
CRAH fan power (kW)	610	290
Rack inlet temperature spread	19-27°C	23-25°C

The annual saving of 7.2 GWh avoids about 2,970 tCO2e.

5. Lessons learned

- Containment first. Raising setpoints before sealing the aisles created hot spots in two rows during the pilot.
- Watch humidity. After the chilled-water temperature was raised, relative humidity in Data Hall 2 rose to 65%; a dedicated dehumidification unit now keeps the dew point within ASHRAE limits.
- Communicate with customers. Two colocation customers queried the higher inlet temperatures; publishing live rack-inlet data against the SLA resolved their concerns.

6. Remaining opportunities

Air-side and chilled-water temperatures are now optimised, but the central chiller plant is still sequenced manually by the operations team using fixed rules, and plant efficiency is only measured monthly. Plant-level optimisation - chiller staging, condenser-water reset and pump and tower-fan speed control - is the next opportunity, but the data-centre team has no in-house precedent for it yet.